SelfieAvatar: High-Fidelity Animatable 3D Head Avatars from In-the-Wild Selfies

Zhen Tian
zhentian.tian@connect.polyu.hk
School of Optometry, The Hong Kong Polytechnic University, Kowloon, Hong Kong
Aimira Innovation, Guangzhou
China

Ziyang Li
ziyang.li.neepu@gmail.com
Westlake University
China

Yuzhen Jiang
sanny@calilia.ai
Aimira Innovation, Guangzhou
University College London, London
University of Toronto, Toronto
Sun Yat-sen University, Guangzhou
China

Huixian Ye
ye_huixian@foxmail.com
Aimira Innovation, Guangzhou
China

Zhuoming Chen
impuuuuuu@gmail.com
Aimira Innovation, Guangzhou
China

Mingguang He✉
mingguang.he@polyu.edu.hk
School of Optometry, The Hong Kong Polytechnic University, Kowloon, Hong Kong
Research Centre for SHARP Vision (RCSV), The Hong Kong Polytechnic University, Kowloon, Hong Kong
Centre for Eye and Vision Research (CEVR), 17W Hong Kong Science Park, Hong Kong.
Hong Kong

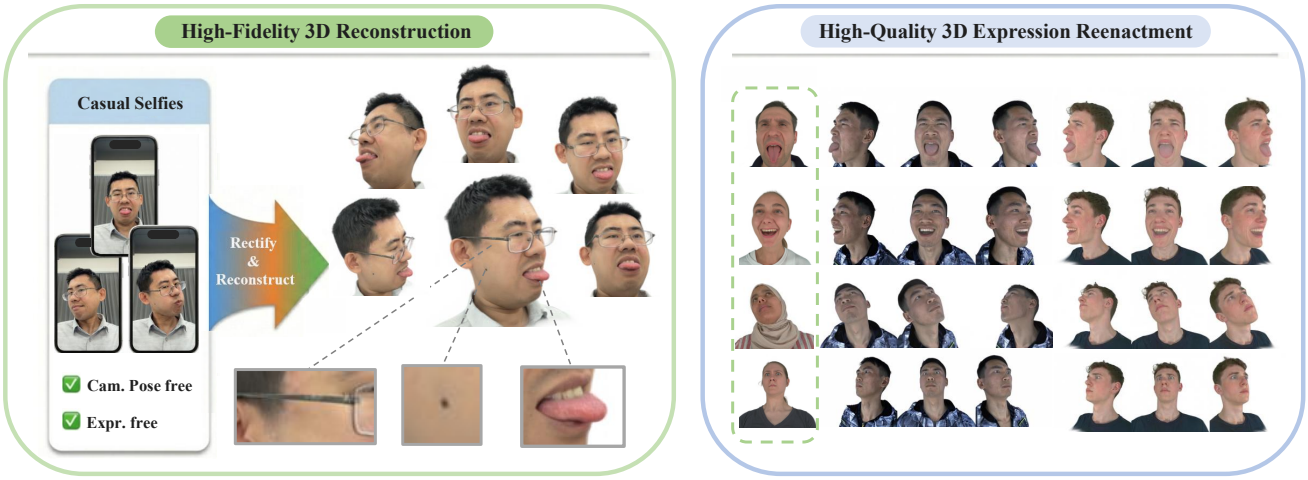


Figure 1: Given three casually captured selfies with unconstrained camera poses and facial expressions, *SelfieAvatar* can reconstruct a high-quality 3D Gaussian head avatar in 10 seconds. Fine-grained details, including lateral nevi, eyeglasses, and non-morphable regions such as the tongue, are faithfully preserved. The reconstructed avatar can be further driven by a image to perform expressive 3D reenactment, producing view-consistent results across diverse head poses and facial expressions.

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Abstract

We present *SelfieAvatar*, a practical system for generating high-fidelity, animatable 3D head avatars from three casually captured, in-the-wild selfies, without requiring camera calibration, fixed facial expressions, or 3D morphable model (3DMM) fitting. While multiple selfies provide stronger geometric cues than single-image input, they also introduce significant challenges due to unknown

camera poses and inconsistent facial expressions across views. To address these issues, we employ a Diffusion Transformer (DiT)-based multi-view rectifier that canonicalizes the input selfies to predefined viewpoints while normalizing them to a shared facial expression. Based on these rectified canonical views, we use a feed-forward Gaussian splatting reconstruction model that produces detailed and geometrically consistent 3D head avatars. Furthermore, we propose a 3DMM-free 3D avatar reenactment framework based on latent diffusion, in which a compact motion embedding extracted from a driving image controls facial expression via cross-attention, and multi-view consistency modules enforce coherent animation across viewpoints. This design enables consistent multi-view facial animation with fine-grained expression preservation, including challenging motions such as tongue movement. Extensive experiments demonstrate that *SelfieAvatar* achieves superior reconstruction quality, view consistency, and animation realism compared to existing methods, highlighting its robustness and practicality for real-world 3D avatar creation from sparse casual inputs.

CCS Concepts

• **Computing methodologies** → **Animation**; **Image-based rendering**; **Point-based models**.

Keywords

Animatable Head Avatar; Single-shot Reconstruction; Few-shot Reconstruction; Gaussian Splatting

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1 Introduction

Photorealistic and animatable 3D head avatars are a key ingredient for immersive telepresence, digital humans, virtual try-on, and interactive entertainment. A practical system should allow casual users to create a high-quality avatar from only a few selfie images, without specialized capture equipment, camera calibration, or lengthy per-subject optimization, while still supporting high-fidelity novel-view rendering and expressive facial animation.

Classical methods for animatable 3D avatar generation typically require constructing avatars from multi-view synchronized captures [41] or single-view video [44, 50, 56, 64] with 3D morphable model (3DMM) parameter tracking, which makes these approaches difficult to deploy for casual users. To overcome these limitations, many recent works explore single-image 3D avatar creation using 3D priors [43, 54, 55, 61], multi-view image or video generation models[45, 46], large reconstruction models [20, 21, 25, 37], as well as image-driven generators [6, 7, 11, 12, 47, 48]. However, single-image reconstruction is underconstrained and often fails to recover reliable side-view geometry. Using a small set of casual multi-view selfies (e.g., left, center, right) provides stronger constraints but introduces challenges due to variations in pose and expression. Although several multi-view methods have been proposed, practical deployment remains challenging. For example, Avat3R [28] requires known camera parameters at inference, while FlexAvatar [39] relies on camera, expression, and pose parameters from 3DMM fitting to achieve high-quality results.

With respect to facial animation and reenactment, many published methods heavily rely on 3DMM for expression control [20, 25, 39]. Although 3DMMs provide low-dimensional control, they limit expressiveness and fail to capture non-morphable details such as tongue motion, and require parameter fitting with external toolchains [18, 40]. In recent years, advances in video generation have driven the development of facial animation control methods, with latent expression control [5, 13, 24, 51] emerging as a representative approach that enables more expressive and stable animations. While effective for 2D video synthesis, extending latent expression control to 3D facial avatars is non-trivial, as 3D representations must simultaneously preserve multi-view coherence across viewpoints, posing additional challenges beyond 2D appearance modeling.

In this work, we present a pipeline for generating high-quality, animatable 3D Gaussian head avatars from only three casual selfies, following a reconstruction-then-animation paradigm. We first employ a DiT-based rectifier to canonicalize camera pose and expression, and reconstruct a 3D head using a feed-forward Gaussian splatting model [57]. Building upon the reconstructed avatar, we then enable animation via an expression-driven multi-view reenactment framework based on latent diffusion, which incorporates identity-preserving reference features and multi-view attention for 3D consistency, and leverages an implicit expression encoder instead of 3DMM-based control.

The principal contributions of this work can be summarized as follows:

- We propose a DiT-based diffusion network that canonicalizes camera pose and facial expression from casual selfies, followed by a large reconstruction model to generate high-quality 3D Gaussian head avatars.
- We introduce a diffusion-based facial reenactment framework with an implicit expression encoder and multi-view cross-attention, enabling expression reenactment with fine-grained details.
- Extensive experiments demonstrate that our model achieves high-fidelity reconstruction and realistic dynamic motion, outperforming prior methods.

2 Related work

2.1 Static 3D Head Reconstruction from sparse views

Research on 3D head reconstruction from sparse views aims to recover a complete 3D head avatar from one or a few images using representations such as meshes, NeRF [36], or Gaussian splatting [23]. Early work largely focused on regressing parameters of classical 3D morphable models (3DMMs) [14, 29, 42, 63], including Basel [38] and FLAME [32]. More recent approaches extend this paradigm with neural 3DMMs to improve geometric fidelity [16, 17, 19]. While these methods are efficient and stable, their reliance on explicit mesh representations limits their ability to model fine-grained appearance details, such as wrinkles, hair, and other non-facial structures.

Another line of work focuses on static 3D head synthesis using 3D-aware generative adversarial networks (GANs), which combine strong generative priors with 3D-aware representations [1, 4, 26, 30, 49]. Although these methods can produce high-quality head appearances near the training distribution, they often struggle to maintain consistent geometry and appearance under large viewpoint variations. In addition, single-image 3D head reconstruction typically relies on latent inversion [8], which frequently leads to degraded reconstruction quality.

More recently, FaceLift [35] introduces a diffusion-based pipeline for static 3D avatar reconstruction, where a multi-view diffusion model [31] synthesizes a consistent set of views from a single image, followed by 3D reconstruction using a separate backbone such as GS-LRM [57]. This formulation substantially improves realism and view consistency over prior pipelines. However, FaceLift lacks explicit multi-view constraints, which often leads to degraded reconstruction quality in lateral views. To address this limitation, we introduce a DiT-based diffusion model that directly operates on multi-view inputs, resulting in improved reconstruction fidelity.

2.2 Animatable 3D Head Reconstruction

Animatable 3D head reconstruction aims to recover a 3D head avatar that supports facial expression control and reenactment. Many prior approaches achieve high-fidelity personalized animatable avatar by fitting 3DMM to single-view or multi-view video sequences and subsequently learning animatable NeRF-based [15, 62, 64] or Gaussian splatting-based [41, 44, 50, 53] head representations driven by the tracked 3DMM motion parameters. While these methods can achieve high-quality results, they rely on time-consuming 3DMM tracking pipelines [40, 63] and typically require per-subject training from scratch, which significantly limits their practicality for real-world use.

To enable animatable avatar generation from a single image or a small number of images, prior work can be broadly classified into four groups: monocular video-based methods, which learn animation from videos using either 3DMM-based expression control [6, 7, 33] or learned expression encoders with synthetic multi-view supervision [11, 12, 48]; diffusion-assisted reconstruction methods, which first synthesize novel views and then perform reconstruction [41, 45, 46, 52]; 3D-prior-based methods, which learn transferable head priors from large-scale 3D datasets and adapt them to new identities through latent identity codes [43, 54, 55, 61]; and canonical head prediction methods, which directly predict a canonical head representation and animate it through pose and expression control, such as position maps [20, 39], expression codes [25, 28], or by directly modifying Gaussian splatting representations with linear blend skinning [21].

Most animatable 3D head models are typically driven by 3DMM parameters, which provide stable control but suffer from limited expressiveness and struggle to capture fine-grained and non-morphable region dynamics. In contrast, latent expression codes have recently emerged as an effective alternative in diffusion-based video avatar generation, enabling more realistic and expressive animation results [5, 34, 59]. In this work, we explore the potential of latent expression control for 3D animatable head avatars.

3 Methodology

3.1 Overview

As illustrated in Figure 2, to reconstruct a 3D avatar from three casual selfies, we first address the domain gap between existing multi-view datasets and in-the-wild selfies by synthesizing selfie-like training images through 3D reconstruction and re-rendering under selfie camera settings (Section 3.2). With this data, we employ a diffusion-based rectifier that takes casual selfies and generates canonical multi-view images with aligned camera viewpoints and a shared facial expression (Section 3.3). These rectified views are then used by a feed-forward Gaussian reconstruction model to produce a detailed 3D Gaussian head avatar (Section 3.4). Finally, for animation, we perform expression-driven reenactment using a diffusion model equipped with an implicit expression latent encoder, which extracts expression cues directly from a single driving image and enables view-consistent, expressive 3D avatar animation (Section 3.5).

3.2 Dataset Preparation

A key challenge in our setting is the domain gap between existing multi-view datasets and casual in-the-wild selfies, particularly in camera intrinsics, viewpoint distributions, and capture distance. To bridge this gap, we synthesize selfie-like training data by reconstructing each frame into a 3D representation and re-rendering it under selfie-style camera configurations.

We start from multi-view image or video datasets, where synchronized images are captured with known camera poses. For each dataset, we train a feed-forward Gaussian reconstruction model, GS-LRM [57], using the provided camera parameters and corresponding images. During training, six images are randomly sampled as inputs, and an additional four images are randomly selected as supervision views. After training, the reconstruction model is used to recover a 3D representation for each frame. We retain only high-quality frames with a peak signal-to-noise ratio (PSNR) exceeding 30 dB. With this representation, frames can be rendered from arbitrary viewpoints.

To simulate casual selfie capture, we render three input views corresponding to left, center, and right selfies. We define the head-centric coordinate system by estimating the scene center from the dataset camera extrinsics and using it as the origin. We fix the camera field of view to 50° to approximate typical front-facing phone cameras. For the left and right views, we uniformly sample azimuth in $[30^\circ, 60^\circ]$ relative to the frontal direction, elevation in $[-15^\circ, 15^\circ]$, and camera radius in $[0.3\text{ m}, 0.5\text{ m}]$. For the center view, we sample a near-frontal configuration with azimuth in $[-5^\circ, 5^\circ]$ and elevation in $[-5^\circ, 5^\circ]$, using a fixed radius of 0.4 m, which reflects typical handheld selfie distances. The rationale behind the chosen camera parameters is detailed in the supplementary material.

In addition, we render a set of fixed viewpoints defined relative to the center camera. These views use azimuth offsets of -60°, -45°, -30°, 30°, 45°, and 60°, while maintaining the same camera radius and elevation as the center view. These canonical views serve as supervision targets for training the rectification model.

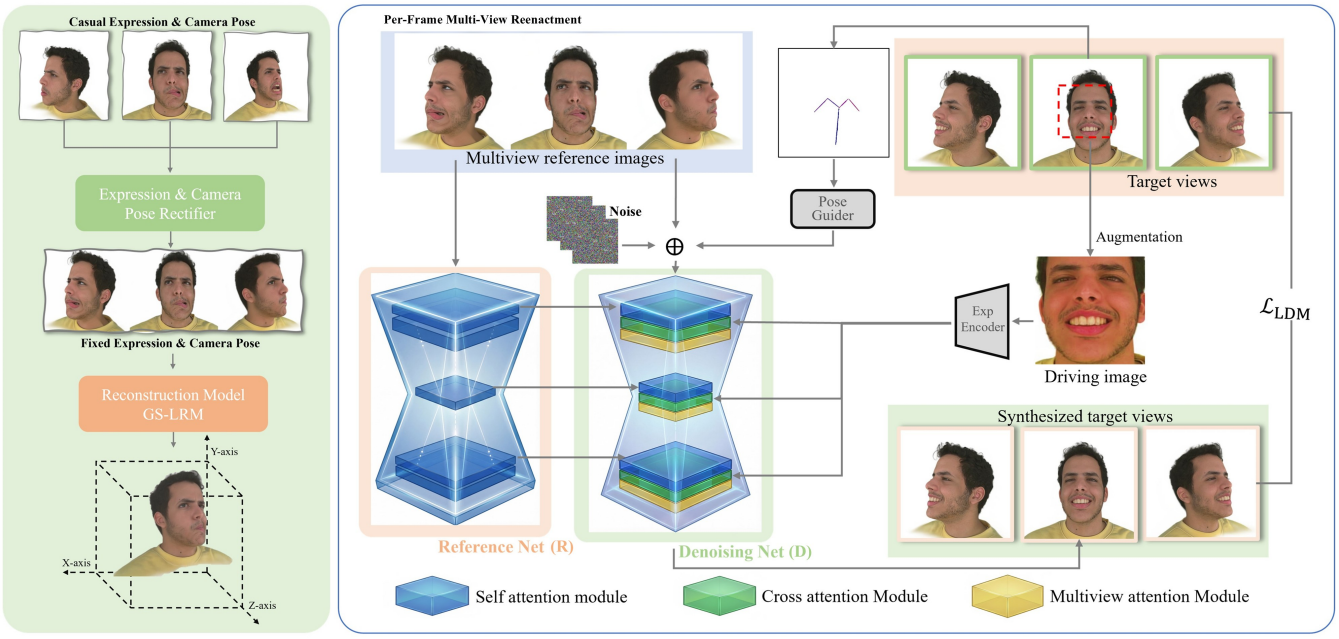


Figure 2: Overview of our framework for 3D head reconstruction and expression reenactment. Given casual selfies, a diffusion-based rectifier generates canonical multi-view images for 3D Gaussian head reconstruction (left). A multi-view diffusion model with implicit expression encoding then enables view-consistent facial reenactment from a single driving image (right).

3.3 Expression and Camera Position Rectifier

Given three casually captured selfie images $\{I_{\text{center}}, I_{\text{left}}, I_{\text{right}}\}$ from different viewpoints, the images may exhibit inconsistent camera poses and facial expressions. Our goal is to transform them into a rectified multi-view set $\{\hat{I}_{\text{center}}, \hat{I}_{\text{left}}, \hat{I}_{\text{right}}\}$ that is aligned to predefined canonical camera viewpoints and maintains expression consistency by anchoring to the center view.

We implement the rectifier as a Diffusion Transformer (DiT) trained under a flow-matching formulation. Specifically, the input images are horizontally concatenated in spatial space to form a composite image: $X_0 = \text{Concat}_{\text{spatial}}(I_{\text{center}}, I_{\text{left}}, I_{\text{right}})$. Following the flow matching formulation, we sample a noise image $\epsilon \sim \mathcal{N}(0, I)$ and a time variable $t \sim \mathcal{U}(0, 1)$, and construct the interpolated state $X_t = (1-t)X_0 + t\epsilon$. A DiT network f_θ is trained to predict the conditional velocity field $v_\theta = f_\theta(X_t, t, c)$, where c denotes a conditioning text prompt specifying the target canonical camera configuration. The predicted velocity defines the probability flow that transports samples from noise to the rectified data distribution.

The rectifier naturally generalizes to flexible input and output configurations. In our experiments, in addition to three-view generation from three input images, we also evaluate three configurations: three-view generation from a single input image, five-view generation from a single input image, and five-view generation from three input images. For single-image input, the center view is triplicated or quintuplicated to construct a multi-view input proxy. For five-view generation from three input views, we construct the input set $\{I_{\text{center}}, I_{\text{left}}, I_{\text{left}}, I_{\text{right}}, I_{\text{right}}\}$ by duplicating the left and right views.

Following FaceLift [35], the outputs always include the origin front view, with $\hat{I}_{\text{center}}$ identical to the center input I_{center}. The target generated left ($\hat{I}_{\text{left}}$) and right ($\hat{I}_{\text{right}}$) views share the same facial expression as the center view (I_{center}), ensuring expression consistency across viewpoints. For three-view outputs, we define azimuth angles at $\{0°, -45°, 45°\}$, whereas for five-view outputs, the angles are set to $\{0°, -30°, -60°, 30°, 60°\}$.

3.4 3D Gaussian Head Reconstruction

We perform sparse-view 3D head reconstruction from the rectified multi-view outputs using GS-LRM [57], a feed-forward model that regresses pixel-aligned 3D Gaussians. It enables high-fidelity reconstruction from sparse views with efficient inference.

We trained the model using either three or five views. For the three-view setting, the input set is $\{\hat{I}_{\text{center}}, \hat{I}_{-45°}, \hat{I}_{45°}\}$, while for the five-view setting, the input set is $\{\hat{I}_{\text{center}}, \hat{I}_{-30°}, \hat{I}_{-60°}, \hat{I}_{30°}, \hat{I}_{60°}\}$. These views correspond to predefined canonical azimuth angles and can be directly generated by the rectification module.

The reconstruction model is trained using a combination of pixel-wise reconstruction loss and perceptual loss. Specifically, the overall training objective is defined as $\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{perc}} \mathcal{L}_{\text{perc}}$, where $\mathcal{L}_{\text{MSE}}$ denotes the mean squared error between images rendered from the reconstructed 3D Gaussians and the corresponding ground-truth images, and $\mathcal{L}_{\text{perc}}$ is a perceptual loss computed using deep image features. At inference time, the reconstruction model directly consumes the rectified multi-view outputs generated by our diffusion-based rectifier and uses predefined camera poses to reconstruct the 3D Gaussian head.

3.5 Expression Transfer

Since GS-LRM enables high-quality 3D reconstruction from sparse multi-view images, we reformulate expression transfer as an expression conditioned multi-view image generation problem. Given a rectified multi-view facial image set of a reference identity, denoted as $\mathcal{I}_R = \{I_R^v\}_{v=1}^n$, our objective is to generate a target multi-view facial image set $\mathcal{I}_{R\to D} = \{I_{R\to D}^v\}_{v=1}^n$, conditioned on a single center-view driving image I_{drv}. The synthesized view set $\mathcal{I}_{R\to D}$ aims to preserve the identity characteristics depicted in the reference set $\mathcal{I}_R$, while accurately reproducing the facial expression and head pose observed in the driving image I_{drv}. All generated views must be 3D-consistent for reliable downstream reconstruction.

Inspired by PersonaLive [34] and X-NeMo [59], we adopt a similar overall structure while adapting it to our multi-view expression transfer setting. As illustrated in Figure 2, the proposed architecture employs a pre-trained diffusion model D as the denoising backbone and a reference network R for identity appearance extraction, together with a pose guider for head pose control and an expression encoder E_{exp} for expression conditioning.

For head pose control, we use a skeleton-based representation encoded by a lightweight MLP pose guider G_{pose} to produce pose features $\mathbf{f}_{\text{pose}}$. To extract expression cues from the driving image (I_{drv}), we first crop the facial region and encode it using an expression encoder E_{exp} into an implicit latent representation $\mathbf{f}_{\text{exp}}$. The encoder is built upon a Face Alignment Network (FAN) backbone [3], augmented by hierarchical self-attention and followed by global pooling and MLP layers that compress expression into a compact 1D latent representation. Following X-NeMo [59], we apply appearance perturbations to the cropped face during training, including scaling, color jittering, gamma augmentation, and partial desaturation, to discourage identity leakage and improve robustness to in-the-wild driving inputs.

The denoising UNet is organized as a stack of repeated blocks, each comprising a spatial processing module, an expression cross-attention module, and a multi-view consistency module. The spatial module performs standard denoising updates on each view's feature map while incorporating pose guidance through $\mathbf{f}_{\text{pose}}$ and injecting reference appearance features from R. Expression control is realized in the expression cross-attention module, where we insert cross-attention layers so that intermediate denoising features attend to the expression embedding $\mathbf{f}_{\text{exp}}$. To explicitly enforce coherence across views, we introduce a multi-view consistency module implemented as a multi-view transformer. Given feature tensors from multiple views, we reshape the features to enable interaction across the view dimension and apply attention to propagate information among views. To reduce memory consumption, we adopt an axial attention strategy [22] that sequentially performs row attention followed by column attention, substantially lowering computational complexity while preserving long-range dependencies. Together, reference-guided appearance conditioning, expression cross-attention and explicit multi-view attention enable the model to generate view-consistent and identity-preserving facial animation results.

During training, we randomly sample two distinct rectified multi-view facial image sets from the same identity, using one set as the reference $\mathcal{I}_R$ and the other as the target set $\mathcal{I}_D$. The center-view of the target set is used as driving image (I_{drv}). The model is trained using the standard latent diffusion noise-prediction objective

$$\mathcal{L}_{LDM} = \mathbb{E}_{t,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t,\ \mathbf{c}_{\text{ref}},\ \mathbf{f}_{\text{exp}},\ \mathbf{f}_{\text{pose}},\ t)\right\|_2^2\right], \tag{1}$$

where $\boldsymbol{\epsilon}_\theta$ denotes the denoising UNet D with parameters θ, $\mathbf{c}_{\text{ref}} = R(\mathcal{I}_R)$ are the reference appearance features extracted by the reference network, $\mathbf{f}_{\text{exp}}$ is the expression latent encoded by E_{exp}, and $\mathbf{f}_{\text{pose}}$ is the pose feature extracted by G_{pose} .

During inference, the rectified multi-view set $\{\hat{I}_{\text{center}}, \hat{I}_{\text{left}}, \hat{I}_{\text{right}}\}$ produced by the rectifier (Section 3.3) is used as the input reference I_R. Together with the target driving image I_{drv}, they are fed into the expression transfer network to generate reenacted multi-view images $I_{R\to D}$. The resulting views are then passed to the 3D reconstruction model (Section 3.4) to obtain the final 3D gaussian representation. For video reenactment, the input sequence is decomposed into frames, and the above pipeline is applied per frame, resulting in a 3D animated sequence.

4 Experiments

4.1 Dataset

We train our model on three datasets: Cafca [2], a synthetic multi-view head dataset, and two real-world multi-view video datasets, NeRSemble (public) [27] and Facescan (private). FaceScan is collected to cover eyeglass-wearing scenarios, where each subject is recorded multiple times with different eyeglass frames. A total of 1,291 subjects are used for training, while 20 real identities from the NeRSemble and Facescan datasets are reserved for evaluation as the test set. All data were uniformly cropped and resized to a resolution of 512x512, and the background was removed using BiRefNet [60].

4.2 Experiments Setup

We evaluate *SelfieAvatar* across diverse 3D head avatar creation scenarios, including static avatar reconstruction from single-view and three-view inputs, and facial expression reenactment. For single-image reconstruction, we compare our method with FaceLift [35], GAGAvatar [6], GPAvatar [7], LAM [21], PanoHead [1], and Portrait4D (referred to as Portrait4D-v2 in the original paper)[12], with a primary focus on reconstruction fidelity. For three-view avatar creation, we compare against GAGAvatar [6], which supports three-view inputs. For self-reenactment and cross-reenactment tasks, we evaluate against GAGAvatar, GPAvatar, LAM, and Portrait4D.

For reconstruction and self-reenactment, where ground-truth images are available, we assess rendering quality using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS) [58]. For identity preservation evaluation, we use cosine similarity of face-recognition embeddings from ArcFace (CSIM) [10] and Average Keypoint Distance (AKD), measured in pixels using landmarks estimated from FaceAlignment [3]. Additionally, we extract expression and pose coefficients from EMOCA [9], a feed-forward network for 3DMM parameter estimation, to compute Average Expression Distance (AED) and Average Pose Distance (APD). For cross-identity reenactment, where ground-truth images are unavailable, we report only CSIM, AED, and APD.

	Model	PSNR↑	SSIM↑	LPIPS↓	CSIM↑	AED↓	APD↓	AKD↓
Single	Facelift	17.618	0.684	0.117	0.857	1.023	0.037	0.082
	GAGAvatar	19.009	0.716	0.112	0.846	1.359	0.039	0.052
	GPAvatar	18.531	0.713	0.108	0.849	1.492	0.027	0.043
	LAM	15.342	0.634	0.149	0.760	1.942	0.052	0.085
	Panohead	14.596	0.562	0.174	0.800	1.508	0.171	0.097
	Portrait4D	18.142	0.698	0.101	0.873	1.335	0.050	0.075
	Ours	**24.667**	**0.835**	**0.038**	**0.968**	**0.465**	**0.011**	**0.031**
Three	GAGAvatar	17.278	0.691	0.170	0.822	1.587	0.040	0.040
	Ours	**24.472**	**0.838**	**0.037**	**0.964**	**0.484**	**0.011**	**0.038**

Table 1: Quantitative comparison of reconstruction quality under single-view and three-view inputs.

View	Input Views	PSNR↑	SSIM↑	LPIPS↓	CSIM↑	AED↓	APD↓	AKD↓
Center	Single View	**24.667**	**0.835**	**0.038**	**0.968**	**0.465**	**0.011**	**0.031**
	Three View	24.472	0.838	0.037	0.964	0.484	0.011	0.038
Left	Single View	18.059	0.685	0.131	0.829	1.056	0.070	0.176
	Three View	**19.698**	**0.731**	**0.098**	**0.887**	**0.942**	**0.062**	**0.208**
Right	Single View	17.811	0.665	0.139	0.816	0.967	0.079	0.307
	Three View	**20.033**	**0.731**	**0.089**	**0.880**	**0.732**	**0.050**	**0.119**
Average	Single View	20.187	0.729	0.102	0.871	0.828	0.053	0.171
	Three View	**21.419**	**0.767**	**0.075**	**0.911**	**0.718**	**0.041**	**0.121**

Table 2: View-wise comparison of our model using single-view and three-view inputs. Three-view inputs consistently improve lateral-view fidelity and overall 3D consistency.

Diffusion Model	PSNR↑	SSIM↑	LPIPS↓	CSIM↑	AED↓	APD↓	AKD↓
Single-view to Three-view	20.187	0.729	0.102	0.871	0.828	0.053	0.171
Single-view to Five-view	18.770	0.706	0.132	0.837	0.994	0.064	0.215
Three-view to Three-view	21.419	0.767	0.075	0.911	0.718	0.041	0.121
Three-view to Five-view	20.479	0.751	0.094	0.889	0.795	0.045	0.232

Table 3: Effect of the number of generated views on reconstruction quality.

4.3 Static 3D Avatar Reconstruction

We randomly select 90 cases from the test set to quantitatively evaluate static avatar creation. Results in Table 1 and Figure 3 demonstrate that our method significantly outperforms other approaches, producing a realistic avatar with excellent 3D consistency in both single-view and three-view avatar creation. As shown in Figure 3 and Table 2, compared to single-view input, our model with three-view input effectively predicts the head shape and preserves lateral details from selfies, resulting in a more accurate and photorealistic 3D avatar. Additional visualizations of the reconstruction results are provided in the supplementary materials.

4.4 Reenactment

We randomly select 100 pairs of reference and target images from the same identity in the test set to evaluate self-reenactment performance. For cross-reenactment, we select 12 representative expressions and 30 reference images from the test subjects, resulting in 360 image pairs for evaluating cross-reenactment capability. The quantitative evaluations are reported in Table 4, while the cross-reenactment results are illustrated in Figure 4. Our model demonstrates superior reenactment performance, effectively capturing fine details including tongue movement, and maintaining strong 3D consistency compared to other methods. More visualizations of reenactment results are provided in the supplementary materials.

4.5 Ablation Studies

Effect of DiT based multi-view rectifier. Previous multi-view generation approaches typically rely on a UNet-based denoising architecture. For example, FaceLift builds upon the UNet backbone of Era3D [31] to synthesize multi-view images. In contrast, our approach employs a DiT-based rectifier, which leverages transformer-style global attention to model long-range spatial dependencies. As demonstrated in Figure 5, the DiT-based rectifier outperforms UNet-based architectures in terms of lateral-view fidelity, leading to more consistent and detailed preservation of facial appearance.

Effect of the Number of Generated Views. Given a single or three input image, the rectifier can generate any number of canonical views. A natural question is whether increasing the number of generated multi-view images further improves reconstruction fidelity. To investigate this, we conduct experiments using three-view and five-view rectified outputs and train the reconstruction model with three and five input views, respectively. As shown in Table 3, increasing the number of multi-view diffusion outputs does not lead to improvement in fidelity when the number of input images is fixed.

Effect of the Multi-view Attention Module. To evaluate the effectiveness of the multi-view consistency module in expression transfer, we compare models trained with and without this component. As shown in Figure 6, removing the multi-view consistency module leads to degraded 3D coherence across views, resulting in blurred boundaries and inconsistent facial details. In contrast, incorporating the multi-view attention module enforces cross-view consistency during expression transfer and significantly improves visual quality and geometric coherence.

4.6 Limitations

Our method is not suitable for real-time animation, as each driving frame requires independent diffusion inference, resulting in a speed of only 0.1 FPS. Moreover, our method does not reconstruct a complete 3D head, particularly in unseen regions such as the back of the head.

5 Conclusions

We presented *SelfieAvatar*, a practical pipeline for generating high-quality, animatable 3D Gaussian head avatars from only three casually captured selfies. Our approach addresses the challenges of in-the-wild few-shot capture with inconsistent camera viewpoints and facial expressions by introducing a DiT-based rectifier that produces canonical multi-view images with aligned camera poses and a shared expression. These rectified views are then fed into a feed-forward Gaussian reconstruction model to recover a detailed 3D head avatar with strong multi-view consistency. Beyond static reconstruction, we propose an expression-latent-driven multi-view reenactment diffusion model with an implicit expression encoder and explicit cross-view attention, enabling expressive and view-consistent animation with fine-grained dynamics. Extensive evaluations demonstrate improved results compared to prior methods.



Figure 3: Qualitative comparison of 3D avatar reconstruction with single-view and three-view inputs on NeRSemble (left column), Facescan (center column), and in-the-wild (right column) selfie data. Our method outperforms prior approaches in fidelity and 3D consistency. Compared to single-view input, three-view input achieves better geometric recovery and preserves more lateral-view details (red boxes). Zoom in for a better view.

	Self reenactment							Cross reenactment		
Model	PSNR↑	SSIM↑	LPIPS↓	CSIM↑	AED↓	APD↓	AKD↓	CSIM↑	AED↓	APD↓
GPAvatar	18.046	0.686	0.141	0.710	2.162	0.047	0.049	0.510	2.745	0.150
GAGAvatar	18.233	0.705	0.112	**0.822**	1.837	0.035	**0.042**	0.640	2.696	0.386
LAM	17.032	0.662	0.144	0.698	2.435	0.046	0.049	0.551	2.830	0.156
Portrait4D	17.984	0.674	0.126	0.760	2.306	0.059	0.065	0.627	2.484	0.149
Ours	**19.581**	**0.732**	**0.091**	0.796	**1.545**	**0.033**	0.050	**0.711**	**2.281**	**0.088**

Table 4: Quantitative evaluation of facial reenactment quality. We evaluate animation performance by transferring facial motion and head pose from a driving image to the same identity (self-reenactment) or to a different identity (cross-reenactment).



Figure 4: Qualitative comparison of cross-identity reenactment results with baselines on NeRSemble (left column), Facescan (center column), and in-the-wild (right column) selfie data. Our method accurately reproduces challenging expressions such as tongue protrusion while maintaining strong 3D consistency across views. In contrast, prior methods often fail or exhibit noticeable artifacts.

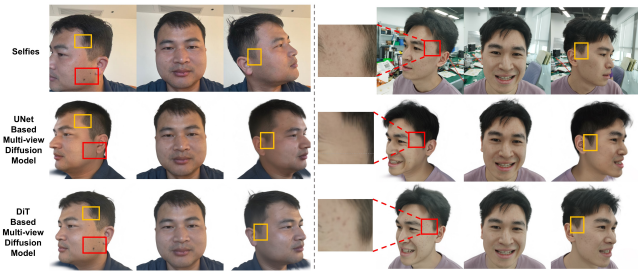


Figure 5: Comparison between UNet- and DiT-based multi-view generation. DiT-based model preserves lateral details like the hairline style (yellow box) and pigmentation (red box) better with three-view input, while UNet-base method tends to generate a more averaged lateral texture.

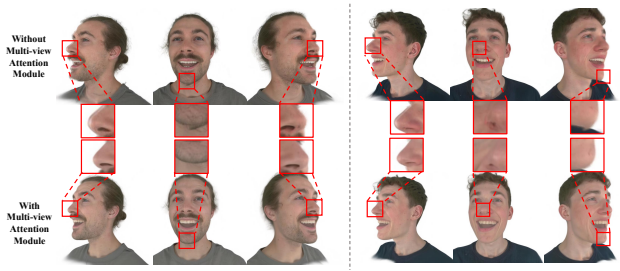


Figure 6: Effect of the multi-view attention module. Multi-view attention improves cross-view consistency and preserves fine facial details across different viewpoints.

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References

[1] Sizhe An, Hongyi Xu, Yichun Shi, Guoxian Song, Umit Y. Ogras, and Linjie Luo. 2023. PanoHead: Geometry-Aware 3D Full-Head Synthesis in 360deg. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 20950–20959.

[2] Marcel C. Buehler, Gengyan Li, Erroll Wood, Leonhard Helminger, Xu Chen, Tanmay Shah, Daoye Wang, Stephan Garbin, Sergio Orts-Escolano, Otmar Hilliges, Dmitry Lagun, Jérémy Riviere, Paulo Gotardo, Thabo Beeler, Abhimitra Meka, and Kripasindhu Sarkar. 2024. Cafca: High-quality Novel View Synthesis of Expressive Faces from Casual Few-shot Captures. In *ACM SIGGRAPH Asia 2024 Conference Paper*. doi:10.1145/3680528.3687580

[3] Adrian Bulat and Georgios Tzimiropoulos. 2017. How far are we from solving the 2D & 3D Face Alignment problem? (and a dataset of 230,000 3D facial landmarks). In *International Conference on Computer Vision*.

[4] Eric R. Chan, Connor Z. Lin, Matthew A. Chan, Koki Nagano, Boxiao Pan, Shalini De Mello, Orazio Gallo, Leonidas Guibas, Jonathan Tremblay, Sameh Khamis, Tero Karras, and Gordon Wetzstein. 2021. Efficient Geometry-aware 3D Generative Adversarial Networks. In *arXiv*.

[5] Gang Cheng, Xin Gao, Li Hu, Siqi Hu, Mingyang Huang, Chaonan Ji, Ju Li, Dechao Meng, Jinwei Qi, Penchong Qiao, Zhen Shen, Yafei Song, Ke Sun, Linrui Tian, Feng Wang, Guangyuan Wang, Qi Wang, Zhongjian Wang, Jiayu Xiao, Sheng Xu, Bang Zhang, Peng Zhang, Xindi Zhang, Zhe Zhang, Jingren Zhou, and Lian Zhuo. 2025. Wan-Animate: Unified Character Animation and Replacement with Holistic Replication. arXiv:2509.14055 [cs.CV] https://arxiv.org/abs/2509.14055

[6] Xuangeng Chu and Tatsuya Harada. 2024. Generalizable and animatable gaussian head avatar. *Advances in Neural Information Processing Systems* 37 (2024), 57642–57670.

[7] Xuangeng Chu, Yu Li, Ailing Zeng, Tianyu Yang, Lijian Lin, Yunfei Liu, and Tatsuya Harada. 2024. GPAvatar: Generalizable and precise head avatar from image (s). *arXiv preprint arXiv:2401.10215* (2024).

[8] Antonia Creswell and Anil Anthony Bharath. 2016. Inverting The Generator Of A Generative Adversarial Network. arXiv:1611.05644 [cs.CV] https://arxiv.org/abs/1611.05644

[9] Radek Daněček, Michael J Black, and Timo Bolkart. 2022. EMOCA: Emotion Driven Monocular Face Capture and Animation. In *CVPR*. 20311–20322.

[10] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. 2019. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4690–4699.

[11] Yu Deng, Duomin Wang, Xiaohang Ren, Xingyu Chen, and Baoyuan Wang. 2024. Portrait4d: Learning one-shot 4d head avatar synthesis using synthetic data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7119–7130.

[12] Yu Deng, Duomin Wang, and Baoyuan Wang. 2024. Portrait4d-v2: Pseudo multi-view data creates better 4d head synthesizer. In *European Conference on Computer Vision*. Springer, 316–333.

[13] Nikita Drobyshev, Antoni Bigata Casademunt, Konstantinos Vougioukas, Zoe Landgraf, Stavros Petridis, and Maja Pantic. 2024. Emoportraits: Emotion-enhanced multimodal one-shot head avatars. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 8498–8507.

[14] Panagiotis P. Filntisis, George Retsinas, Foivos Paraperas-Papantoniou, Athanasios Katsamanis, Anastasios Roussos, and Petros Maragos. 2023. SPECTRE: Visual Speech-Informed Perceptual 3D Facial Expression Reconstruction From Videos. In *CVPRW*. 5745–5755.

[15] Guy Gafni, Justus Thies, Michael Zollhöfer, and Matthias Nießner. 2021. Dynamic Neural Radiance Fields for Monocular 4D Facial Avatar Reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 8649–8658.

[16] Simon Giebenhain, Tobias Kirschstein, Markos Georgopoulos, Martin Rünz, Lourdes Agapito, and Matthias Nießner. 2023. Learning neural parametric head models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 21003–21012.

[17] Simon Giebenhain, Tobias Kirschstein, Markos Georgopoulos, Martin Rünz, Lourdes Agapito, and Matthias Nießner. 2024. MonoNPHM: Dynamic Head Reconstruction from Monocular Videos. In *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*.

[18] Simon Giebenhain, Tobias Kirschstein, Rünz Rünz, Lourdes Agapito, and Matthias Nießner. 2025. Pixel3DMM: Versatile Screen-Space Priors for Single-Image 3D Face Reconstruction. *arXiv preprint arXiv:2505.00615* (2025).

[19] Simon Giebenhain, Tobias Kirschstein, Liam Schoneveld, Davide Davoli, Zhe Chen, and Matthias Nießner. 2025. Pix2NPHM: Learning to Regress NPHM Reconstructions From a Single Image. arXiv:2512.17773 [cs.CV] https://arxiv.org/abs/2512.17773

[20] Chen Guo, Zhuo Su, Jian Wang, Shuang Li, Xu Chang, Zhaohu Li, Yang Zhao, Guidong Wang, and Ruqi Huang. 2025. SEGA: Drivable 3D Gaussian Head Avatar from a Single Image. *arXiv preprint arXiv:2504.14373* (2025).

[21] Yisheng He, Xiaodong Gu, Xiaodan Ye, Chao Xu, Zhengyi Zhao, Yuan Dong, Weihao Yuan, Zilong Dong, and Liefeng Bo. 2025. LAM: Large Avatar Model for One-shot Animatable Gaussian Head. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*. 1–13.

[22] Jonathan Ho, Nal Kalchbrenner, Dirk Weissenborn, and Tim Salimans. 2019. Axial Attention in Multidimensional Transformers. arXiv:1912.12180 [cs.CV] https://arxiv.org/abs/1912.12180

[23] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (July 2023). https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/

[24] Taekyung Ki, Dongchan Min, and Gyeongsu Chae. 2024. Float: Generative motion latent flow matching for audio-driven talking portrait. *arXiv preprint arXiv:2412.01064* (2024).

[25] Tobias Kirschstein, Simon Giebenhain, and Matthias Nießner. 2025. FlexAvatar: Learning Complete 3D Head Avatars with Partial Supervision. arXiv:2512.15599 [cs.CV] https://arxiv.org/abs/2512.15599

[26] Tobias Kirschstein, Simon Giebenhain, Jiapeng Tang, Markos Georgopoulos, and Matthias Nießner. 2024. GGHead: Fast and Generalizable 3D Gaussian Heads. *arXiv preprint arXiv:2406.09377* (2024).

[27] Tobias Kirschstein, Shenhan Qian, Simon Giebenhain, Tim Walter, and Matthias Nießner. 2023. NeRSemble: Multi-View Radiance Field Reconstruction of Human Heads. *ACM Trans. Graph.* 42, 4, Article 161 (jul 2023), 14 pages. doi:10.1145/3592455

[28] Tobias Kirschstein, Javier Romero, Artem Sevastopolsky, Matthias Nießner, and Shunsuke Saito. 2025. Avat3r: Large Animatable Gaussian Reconstruction Model for High-fidelity 3D Head Avatars. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 12089–12100.

[29] Chunlu Li, Andreas Morel-Forster, Thomas Vetter, Bernhard Egger, and Adam Kortylewski. 2021. To fit or not to fit: Model-based Face Reconstruction and Occlusion Segmentation from Weak Supervision. *CoRR* abs/2106.09614 (2021). arXiv:2106.09614 https://arxiv.org/abs/2106.09614

[30] Heyuan Li, Ce Chen, Tianhao Shi, Yuda Qiu, Sizhe An, Guanying Chen, and Xiaoguang Han. 2024. SphereHead: Stable 3D Full-head Synthesis with Spherical Tri-plane Representation. arXiv:2404.05680 [cs.CV]

[31] Peng Li, Yuan Liu, Xiaoxiao Long, Feihu Zhang, Cheng Lin, Mengfei Li, Xingqun Qi, Shanghang Zhang, Wenhan Luo, Ping Tan, et al. 2024. Era3D: High-Resolution Multiview Diffusion using Efficient Row-wise Attention. *arXiv preprint arXiv:2405.11616* (2024).

[32] Tianye Li, Timo Bolkart, Michael J Black, Hao Li, and Javier Romero. 2017. Learning a model of facial shape and expression from 4D scans. *ACM Trans. Graph.* 36, 6 (2017), 194–1.

[33] Xueting Li, Shalini De Mello, Sifei Liu, Koki Nagano, Umar Iqbal, and Jan Kautz. 2023. Generalizable one-shot 3d neural head avatar. *Advances in Neural Information Processing Systems* 36 (2023), 47239–47250.

[34] Zhiyuan Li, Chi-Man Pun, Chen Fang, Jue Wang, and Xiaodong Cun. 2025. PersonaLive! Expressive Portrait Image Animation for Live Streaming. *arXiv preprint arXiv:2512.11253* (2025).

[35] Weijie Lyu, Yi Zhou, Ming-Hsuan Yang, and Zhixin Shu. 2025. FaceLift: Learning Generalizable Single Image 3D Face Reconstruction from Synthetic Heads. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 12691–12701.

[36] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2021. NeRF: representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (Dec. 2021), 99–106. doi:10.1145/3503250

[37] Antonio Oroz, Matthias Nießner, and Tobias Kirschstein. 2025. PercHead: Perceptual Head Model for Single-Image 3D Head Reconstruction Editing. arXiv:2511.02777 [cs.CV] https://arxiv.org/abs/2511.02777

[38] Pascal Paysan, Reinhard Knothe, Brian Amberg, Sami Romdhani, and Thomas Vetter. 2009. A 3D face model for pose and illumination invariant face recognition. In *2009 sixth IEEE international conference on advanced video and signal based surveillance*. Ieee, 296–301.

[39] Cheng Peng, Zhuo Su, Liao Wang, Chen Guo, Zhaohu Li, Chengjiang Long, Zheng Lv, Jingxiang Sun, Chenyangguang Zhang, and Yebin Liu. 2025. FlexAvatar: Flexible Large Reconstruction Model for Animatable Gaussian Head Avatars with Detailed Deformation. arXiv:2512.17717 [cs.CV] https://arxiv.org/abs/2512.17717

[40] Shenhan Qian. 2024. VHAP: Versatile Head Alignment with Adaptive Appearance Priors. doi:10.5281/zenodo.14988309

[41] Shenhan Qian, Tobias Kirschstein, Liam Schoneveld, Davide Davoli, Simon Giebenhain, and Matthias Nießner. 2024. Gaussianavatars: Photorealistic head avatars with rigged 3d gaussians. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20299–20309.

[42] George Retsinas, Panagiotis P. Filntisis, Radek Danecek, Victoria F. Abrevaya, Anastasios Roussos, Timo Bolkart, and Petros Maragos. 2024. 3D Facial Expressions through Analysis-by-Neural-Synthesis. In *Conference on Computer Vision and Pattern Recognition (CVPR)*.

[43] Jack Saunders, Charlie Hewitt, Yanan Jian, Marek Kowalski, Tadas Baltrusaitis, Yiye Chen, Darren Cosker, Virginia Estellers, Nicholas Gydé, Vinay P Namboodiri, et al. 2025. GASP: Gaussian Avatars with Synthetic Priors. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 271–280.

[44] Zhijing Shao, Zhaolong Wang, Zhuang Li, Duotun Wang, Xiangru Lin, Yu Zhang, Mingming Fan, and Zeyu Wang. 2024. SplattingAvatar: Realistic Real-Time Human Avatars with Mesh-Embedded Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[45] Felix Taubner, Ruihang Zhang, Mathieu Tuli, Sherwin Bahmani, and David B Lindell. 2025. MVP4D: Multi-View Portrait Video Diffusion for Animatable 4D Avatars. *arXiv preprint arXiv:2510.12785* (2025).

[46] Felix Taubner, Ruihang Zhang, Mathieu Tuli, and David B Lindell. 2025. Cap4d: Creating animatable 4d portrait avatars with morphable multi-view diffusion models. In *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE Computer Society, 5318–5330.

[47] Phong Tran, Egor Zakharov, Long-Nhat Ho, Liwen Hu, Adilbek Karmanov, Aviral Agarwal, McLean Goldwhite, Ariana Bermudez Venegas, Anh Tuan Tran, and Hao Li. 2024. VOODOO XP: Expressive One-Shot Head Reenactment for VR Telepresence. *ACM Transactions on Graphics, Proceedings of the 17th ACM SIGGRAPH Conference and Exhibition in Asia 2024, (SIGGRAPH Asia 2024), 12/2024* (2024).

[48] Phong Tran, Egor Zakharov, Long-Nhat Ho, Anh Tuan Tran, Liwen Hu, and Hao Li. 2024. VOODOO 3D: Volumetric Portrait Disentanglement for One-Shot 3D Head Reenactment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

[49] Alex Trevithick, Matthew Chan, Michael Stengel, Eric R. Chan, Chao Liu, Zhiding Yu, Sameh Khamis, Manmohan Chandraker, Ravi Ramamoorthi, and Koki Nagano. 2023. Real-Time Radiance Fields for Single-Image Portrait View Synthesis. In *ACM Transactions on Graphics (SIGGRAPH)*.

[50] Jun Xiang, Xuan Gao, Yudong Guo, and Juyong Zhang. 2024. FlashAvatar: High-fidelity Head Avatar with Efficient Gaussian Embedding. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[51] Sicheng Xu, Guojun Chen, Yu-Xiao Guo, Jiaolong Yang, Chong Li, Zhenyu Zang, Yizhong Zhang, Xin Tong, and Baining Guo. 2024. Vasa-1: Lifelike audio-driven talking faces generated in real time. *Advances in Neural Information Processing Systems* 37 (2024), 660–684.

[52] Sicheng Xu, Guojun Chen, Jiaolong Yang, Yizhong Zhang, Yu Deng, Stephen Lin, and Baining Guo. 2025. VASA-3D: Lifelike Audio-Driven Gaussian Head Avatars from a Single Image. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.

[53] Yuelang Xu, Benwang Chen, Zhe Li, Hongwen Zhang, Lizhen Wang, Zerong Zheng, and Yebin Liu. 2024. Gaussian Head Avatar: Ultra High-fidelity Head Avatar via Dynamic Gaussians. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

[54] Yuelang Xu, Lizhen Wang, Zerong Zheng, Zhaoqi Su, and Yebin Liu. 2024. 3d gaussian parametric head model. In *European Conference on Computer Vision*. Springer, 129–147.

[55] Zhixuan Yu, Ziqian Bai, Abhimitra Meka, Feitong Tan, Qiangeng Xu, Rohit Pandey, Sean Fanello, Hyun Soo Park, and Yinda Zhang. 2024. One2Avatar: Generative Implicit Head Avatar For Few-shot User Adaptation. *arXiv preprint arXiv:2402.11909* (2024).

[56] Jiawei Zhang, Zijian Wu, Zhiyang Liang, Yicheng Gong, Dongfang Hu, Yao Yao, Xun Cao, and Hao Zhu. 2025. Fate: Full-head gaussian avatar with textural editing from monocular video. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 5535–5545.

[57] Kai Zhang, Sai Bi, Hao Tan, Yuanbo Xiangli, Nanxuan Zhao, Kalyan Sunkavalli, and Zexiang Xu. 2024. GS-LRM: Large Reconstruction Model for 3D Gaussian Splatting. *European Conference on Computer Vision* (2024).

[58] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. 2018. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 586–595.

[59] Xiaochen Zhao, Hongyi Xu, Guoxian Song, You Xie, Chenxu Zhang, Xiu Li, Linjie Luo, Jinli Suo, and Yebin Liu. 2025. X-NeMo: Expressive neural motion reenactment via disentangled latent attention. *arXiv preprint arXiv:2507.23143* (2025).

[60] Peng Zheng, Dehong Gao, Deng-Ping Fan, Li Liu, Jorma Laaksonen, Wanli Ouyang, and Nicu Sebe. 2024. Bilateral Reference for High-Resolution Dichotomous Image Segmentation. *CAAI Artificial Intelligence Research* 3 (2024), 9150038.

[61] Xiaozheng Zheng, Chao Wen, Zhaohu Li, Weiyi Zhang, Zhuo Su, Xu Chang, Yang Zhao, Zheng Lv, Xiaoyuan Zhang, Yongjie Zhang, et al. 2025. Headgap: Few-shot 3d head avatar via generalizable gaussian priors. In *2025 International Conference on 3D Vision (3DV)*. IEEE, 946–957.

[62] Yufeng Zheng, Victoria Fernández Abrevaya, Marcel C. Bühler, Xu Chen, Michael J. Black, and Otmar Hilliges. 2022. I M Avatar: Implicit Morphable Head Avatars from Videos. In *Computer Vision and Pattern Recognition (CVPR)*.

[63] Wojciech Zielonka, Timo Bolkart, and Justus Thies. 2022. Towards Metrical Reconstruction of Human Faces. In *European Conference on Computer Vision*. 250–269.

[64] Wojciech Zielonka, Timo Bolkart, and Justus Thies. 2023. Instant Volumetric Head Avatars. In *Conference on Computer Vision and Pattern Recognition*.